

Infrastructure

Version: 1.0

Purpose

This document describes the Infrastructure layer of Trading Platform Pro.

The Infrastructure layer provides the technical capabilities required by the application while remaining isolated from the business domain.

It contains implementations of interfaces defined by the Domain and Application layers.

Responsibilities

Infrastructure is responsible for:

- Configuration
- Logging
- Dependency Injection
- Persistence
- Messaging
- File System
- Runtime Services
- Scheduling
- Monitoring
- Health Checks
- Serialization
- Plugin Management
- External APIs
- Broker Integration

Design Principles

Infrastructure shall:

- implement interfaces
- remain replaceable
- contain no business rules
- isolate external dependencies
- support automated testing

Business decisions always belong to the Domain.

Infrastructure Packages

Typical packages include:

```

```
bootstrap/
cache/
clock/
config/
files/
health/
identity/
logging/
messaging/
monitoring/
plugins/
resources/
```

runtime/  
scheduler/  
security/  
serialization/  
transactions/  
...

Each package has a single technical responsibility.

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## External Dependencies

Infrastructure communicates with external systems such as:

- File System
- Databases
- Broker APIs
- Market Data Providers
- Operating System
- Network Services

These dependencies are isolated from the Domain.

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## Dependency Injection

Infrastructure services are registered during application startup.

Service creation is centralized to:

- simplify testing
- improve maintainability
- avoid hidden dependencies

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## Persistence

Persistence implementations belong exclusively to Infrastructure.

Typical responsibilities:

- repositories
- file storage
- configuration loading
- serialization

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## Messaging

Messaging enables communication between components without introducing tight coupling.

Examples:

- domain events
- application events
- notifications

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## Runtime Services

Infrastructure provides shared runtime functionality such as:

- clocks
- schedulers
- monitoring
- health checks
- diagnostics

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## Testing

Infrastructure components should be:

- independently testable
- mockable
- deterministic where possible

External systems should always be abstracted behind interfaces.

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## Evolution

New infrastructure modules should:

- remain loosely coupled
- follow existing package conventions
- avoid unnecessary dependencies
- preserve architectural consistency

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## Infrastructure Boundaries

Infrastructure provides technical capabilities only.

It must never contain:

- business rules
- trading decisions
- domain validation
- business workflows

Business logic belongs exclusively to the Domain and Application layers.

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## Configuration Strategy

Configuration shall:

- support multiple environments
- remain externalized
- be validated during startup
- never contain secrets in source control

Use dedicated providers for secrets and credentials.

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## Resilience

Infrastructure components should:

- fail fast
- support graceful recovery
- provide meaningful diagnostics
- expose health information
- degrade gracefully where appropriate

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## Infrastructure Review Checklist

Before merging verify:

- no business logic introduced
- interfaces respected
- dependencies minimized
- external services abstracted
- deterministic behaviour preserved
- documentation updated

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## Related Documents

- Architecture.md

- Project\_Structure.md
- Domain\_Model.md
- AGENTS.md